



Google Summer of Code - 2022

ScummVM Macromedia Director

19 April 2022

Pragyansh Chaturvedi

B. Tech CSE '24

Indian Institute of Technology, Roorkee

Uttarakhand, India

About Me

Personal Info

- Name : Pragyansh Chaturvedi
- Email : pragyanshchaturvedi18@gmail.com
- Github : <https://github.com/r41k0u>
- Discord Name : Raikou#9980
- Mentor : Eugene Sandulenko
- Location : India
- Timezone : IST (UTC +05:30)

Educational Info

- University : Indian Institute of Technology, Roorkee
- Major : Computer Science and Engineering
- Current Year : Sophomore (2nd) (Expected Graduation : 2024)
- Degree : Bachelor of Technology (B.Tech)

Background

I am Pragyansh Chaturvedi, UG CSE Sophomore at Indian Institute of Technology, Roorkee. I have been involved in various software development and tech-related endeavors in the form of hackathons, CTFs, course projects, and projects at SDS Labs (a student technical group at IITR I am a part of)

I attended the EEN-103 (Fundamentals of C++), CSN-291 (Object-Oriented Analysis and Design), CSN-232 (Operating Systems), and CSN-252 (System Software) courses at my university. These are the courses I was most interested in.



I primarily code in C++ and Python. I am pretty familiar with the various characteristics of C++ like pointers, inheritance, polymorphism, etc.

Programming Experience

- Languages: C++, C, Lua, Python, Rust, Java, Javascript, C#
- Software Packages: Git, Ghidra, IDA, GDB, ReStructuredText, Docker, Figma, Vim
- Miscellaneous: Object-Oriented Programming, Analysis, and Design

Past Projects / Experience

Rootex

- Rootex is an advanced C++ 3D game engine powering an in-production game yet to be announced. The game will finally ship on Windows and use DirectX 11. The engine is based on the Entity-Component-System architecture and uses a Lua scripting Engine for making games.
[GitHub](#) | [Docs](#)
- My contribution to Rootex covers creating the in-development game to be shipped with Rootex, fixing bugs in the engine, and tweaking it for the game.
- The engine was also covered by GamesFromScratch ([Link](#))

CTFs and Reversing

I regularly participate in ongoing Capture the Flag (CTF) events: A time-limited cybersecurity event where challenges are given which must be exploited to obtain a readable ASCII flag.

I am a member of the teams SDSLabs and InfosecIIITR.

Team SDSLabs was placed 2nd in India and 6th among UG World teams in the CSAW 2021 CTF Finals, organized by New York University. I was a part of the same.

My main areas of interest are reverse engineering and binary exploitation.

Here is a recent reverse engineering challenge I solved in UTCTF 2022.

UTCTF Adventure ROM 4

[File](#)



[Writeup](#). This was a recent basic reversing challenge I documented. Though very basic reversing is done without using Ghidra or IDA, I do most of the challenges using them, like the Flare On challenge.

Outrun Chase

Outrun Chase is an Outrun/Retrowave-themed racing/shooting game made in the Godot game engine that provides two game modes for the players to compete in, racing and deathmatch. [Link](#)

Vibe

Vibe is a musical rhythm-based game created as an entry for the Global Game Jam 2022. Made on the Godot Game Engine. [Github](#) | [Game](#)

PatentNFT

Incorporating Cryptographic Commitment Schemes using Non Fungible Tokens on the Solana network to mitigate the problem of intellectual property ownership conflicts.

This project was also a winner at Solana's Building Out Loud hackathon. [GitHub](#) | [Submission](#) | [Link](#)



Da'Chess

Da'Chess is a 3D PvP Chess playing, practicing, and streaming platform that allows you to stake tokens on games, bet on players or tip your favorite streamers on the Polygon Network. [GitHub-F](#) | [GitHub-B](#) | [Link](#)

Contributions

To explore the various components of the ScummVM project and get familiar with the codebase, I made some contributions to the project. I tried to examine multiple components rather than restricting myself to 1 or 2 only and focused on learning lots about the component rather than just fixing the bug.

Issues

- While working on fixing a bug in the NGL Engine, I came across the following bug: [Trac #13270](#)
- I had opened this issue too regarding the debugger, thinking that this was not the intended functionality. I was corrected by criezy later, but I got to know about the debugger, FSNode, Path, and SearchMan by this: [Trac #13343](#)

Pull Requests

- [Trac #9669](#) | [Pull #3662](#) | Merged
My First Pull Request. The function to display the cactus base in scene 31 was missing when the cactus state was changed. The same was implemented. This was my initiation to the ScummVM codebase and gave me an elementary idea of NGL scenes.
- [Trac #10395](#) | [Pull #3701](#) | Merged
The player would enter the scene at a blocked entrance and couldn't move. So changed the entrance based on whether the Foot is in place. I learned about scene transitions in NGL and the message-queue framework, which is its core functioning.
- [Trac #10396](#) | [Pull #3713](#) | Merged
The engine waited for the animation to get over before changing the

bridge state. This change was very small. I found [Trac #13270](#) while working on this.

- [Trac #13196](#) | [Pull #3731](#) | Merged

My next PR was on the SCUMM engine. I learned a lot about the game detection system and the SCUMM engine. The SCUMM engine has its various opcodes and a stack that is executed in the main loop. SCUMM Engine functions as an actual VM. I logged the opcodes, objects, and actors during the bug's reproduction and analyzed the logs. This gave me an idea of how scripts and commands work. A brief of my findings is [here](#).

- [Trac #13343](#) | [Pull #3745](#) | Closed without merging

While fiddling with the debugger, I thought I had found a bug. I reported it and made a Pull Request to fix it. But it turned out to be working as intended and was not a bug. This was closed without merging, but I learned about the debugger while working on this.

- [Trac #11386](#) | [Pull #3746](#) | Merged (in another commit)

This Pull Request implemented commandline options to find the MD5 hash of the game file for easy reporting of unknown games. I got familiar with Plugins, MetaEngineDetection and AdvancedMetaEngineDetection. During this PR, I had inherited the ScummMetaEngineDetection from AdvancedMetaEngineDetection instead of MetaEngineDetection, but reverted the change and added a check instead.

- [Pull #3768](#) | Merged

This Pull Request was for implementing the STUBbed builtin function **erase cast** in the Director Engine. This was my initiation to the Director Engine, and I got to know about the basic framework (movie, channels, frames, sprites, cast)

- [Pull #3773](#) | Merged

This was my next contribution to the Director Engine. Implemented the **duplicate cast** lingo command in the builtins.

Project

Macromedia Director

Macromedia Director was a media application that allowed users to create interactive movies/games based on the movie metaphor (like an actual reel).

It could incorporate many different bitmap, audio, and video file formats, making it possible to integrate media.

It used the **Lingo** scripting language to manipulate audio and sprites and other things like channels, frames, and castMembers.

Various functionalities that were not built into Lingo could be used by adding **Xtras**. An XDK (Xtra Development Kit) was provided to create C++ based Xtras, which could be used for File I/O, hardware calls, and advanced multimedia functionality.

The movies created could be exported to the **.dcr**, which could run on the shockwave plugin, or a binary executable could also be exported, which can run natively on systems.

A large number of games were created using the Macromedia Director, ScummVM provides a [list](#) for the games detected correctly and can run to some extent.

Structure

The two most important constructs of the Macromedia Director are the **Cast** and the **Score**.

The Cast is a collection of **CastMembers**. Each CastMember can have artwork like sprites, sounds, and colors contained in them which can be used in the movie.



The Score consists of **Channels** and **Frames**. Channels are containers in the Score that can contain a CastMember, transition, or Lingo Command over the progression of time. Frames tell the state of the movie at a given time. The horizontal rows in a score are channels, and the vertical columns are frames. Their intersection creates cells.

A good primer for the Macromedia Director is [this](#).

Tasks

I will be completing the following tasks during the Google Summer of Code 2022:

- Workshop Movies: Completing support for all the workshop movies.
- STUBbed Code: Implementing all the stubbed functions
- Meet Mediaband: Finish support for Meet Mediaband
- The Journeyman Project: Add support for The Journeyman Project in the Director Engine

I would be focusing on support for Director 4 in the Director Engine and testing it out.

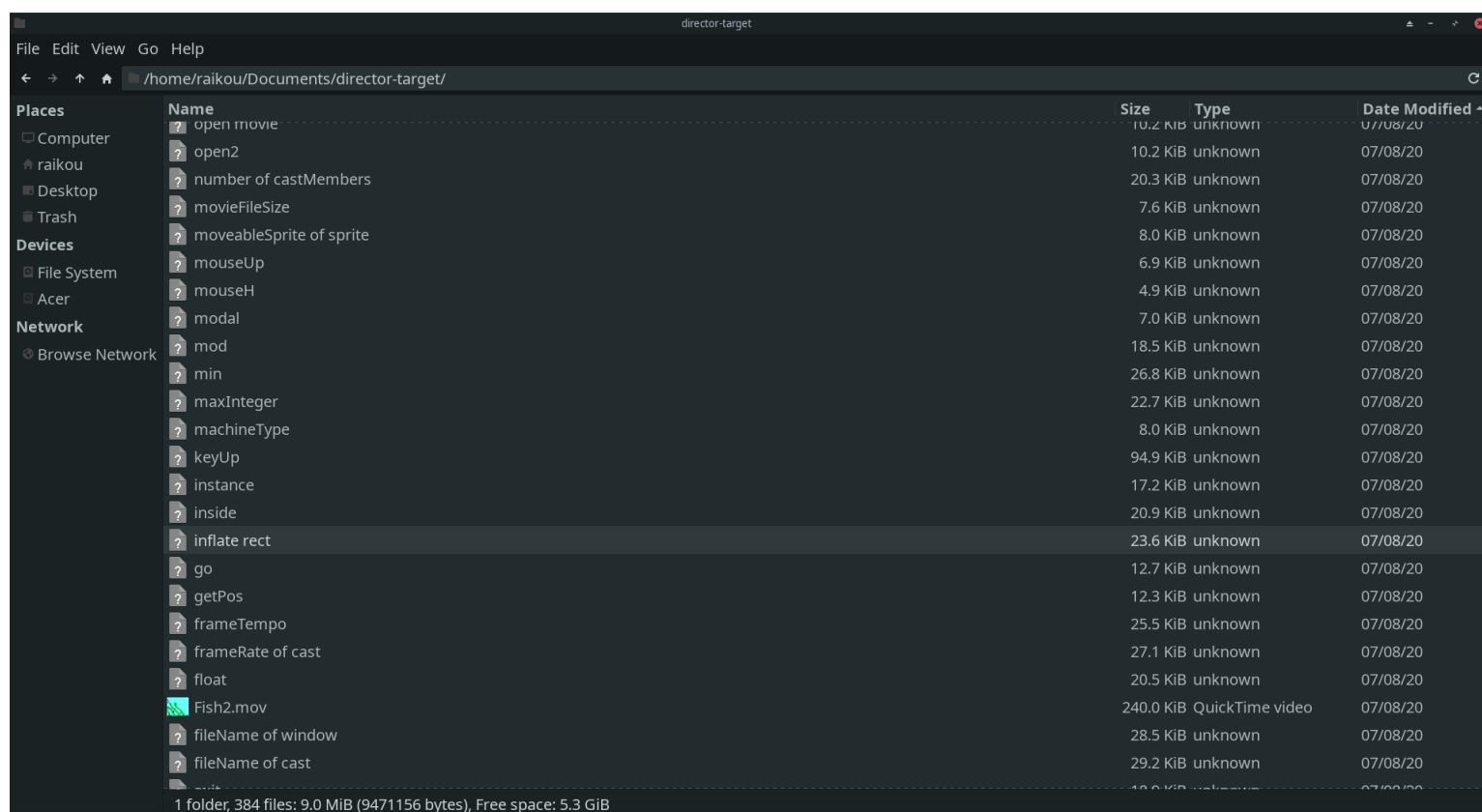
Stretch Goal:

- Extend Support for Director 5 by implementing Director 5 lingo commands and making Workshop movies for the same. I would focus on this after my primary tasks have been fulfilled and there is still time in the GSoC timeline.

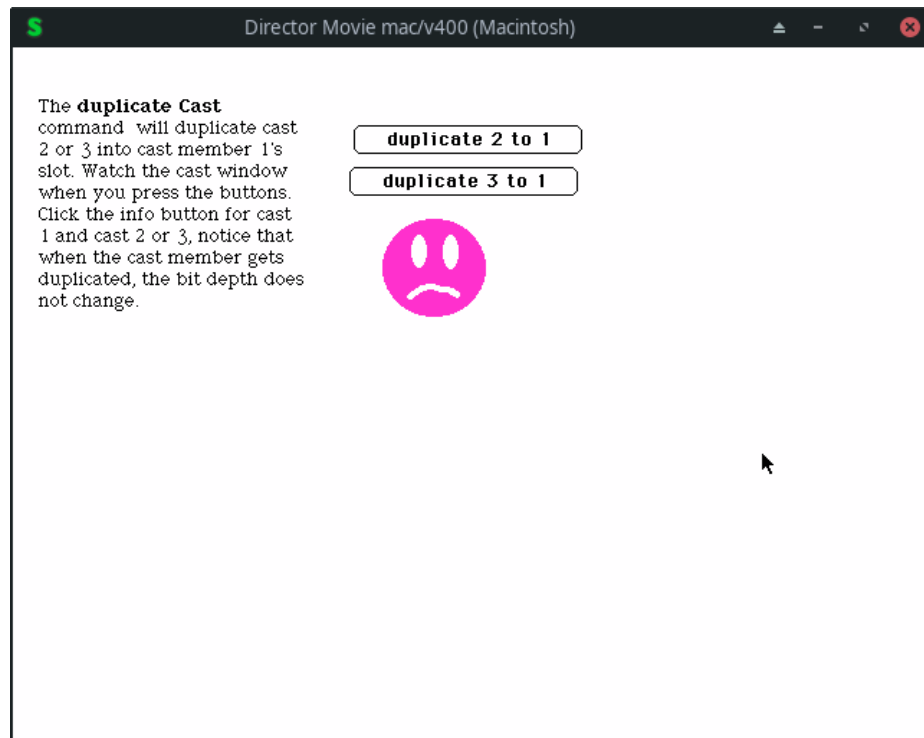
Workshop Movies

Workshop movies is a collection of 370 short movies to test the various functionalities of the Director Engine. Most of them are tests for some lingo commands.

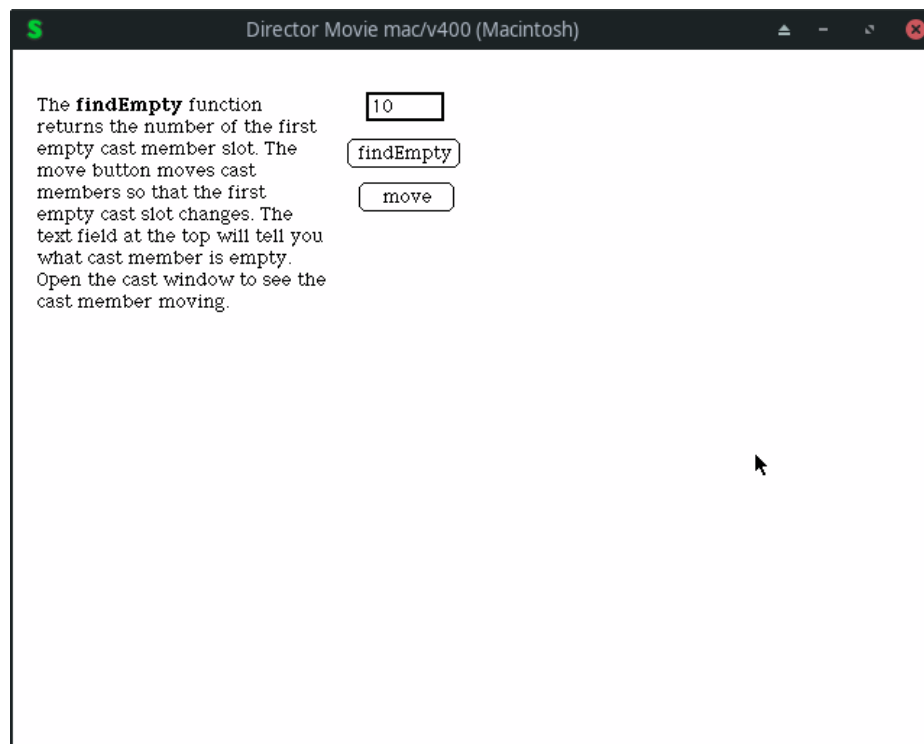
A lot of them don't have their functionalities implemented. My first task would be to implement the functionalities of all the workshop movies. This is elaborated more in the next section.



(A view of some workshop movies)



(the "duplicate cast" workshop movie)



("findEmpty" workshop movie, which is not functional right now)

STUBbed code

The Director Engine's code has 277 instances of "stub" in the codebase. The stubbed code is mostly in the xlibs, lingo-the.cpp, and lingo-the.cpp

Following is an example of a STUBbed code. director/lingo/lingo-builtins.cpp, b_open(), line 1107:

```
void LB::b_open(int nargs) {
    g_lingo->printSTUBWithArglist("b_open", nargs);

    g_lingo->dropStack(nargs);
}
```

printSTUBWithArglist():

```
void Lingo::printSTUBWithArglist(const char *funcname, int nargs, const char *prefix) {
    Common::String s(funcname);

    s += '(';

    for (int i = 0; i < nargs; i++) {
        Datum d = _stack[_stack.size() - nargs + i];

        s += d.asString(true);

        if (i != nargs - 1)
            s += ", ";
    }

    s += ")";

    debug(5, "%s %s", prefix, s.c_str());
}
```

These STUBs have to be implemented as these lingo commands are not being executed right now, due to which many director games and movies run with missing/broken functionality in ScummVM.

Here is a way in which I will be doing this. Following is the `b_erase` function I unSTUBbed:

Before:

```
void LB::b_erase(int nargs) {
    g_lingo->printSTUBWithArglist("b_erase", nargs);

    g_lingo->dropStack(nargs);
}
```

After:

```
void LB::b_erase(int nargs) {
    Datum d = g_lingo->pop();
    CastMember *eraseCast = g_director->getCurrentMovie()->getCastMember(d.asMemberID());

    if (eraseCast) {
        eraseCast->_erase = true;
        Common::Array<Channel *> channels = g_director->getCurrentMovie()->getScore()->_channels;

        for (uint i = 0; i < channels.size(); i++) {
            if (channels[i]->_sprite->_castId == d.asMemberID()) {
                channels[i]->_dirty = true;
            }
        }
    }
}
```

To do this, I first looked up the command in the Lingo Dictionary. Then looked at the relevant parts of code that would be affected by this functionality. Then did the required changes. My last 2 PRs are on implementing STUB code, [Pull #3768](#) and [Pull #3773](#).



Implementing the STUB code will also complete the Workshop Movies, as the non-functional movies test STUB commands. This would also help with The Journeyman Project target, discussed later.

Meet Mediaband

Meet Mediaband is one target that uses Macromedia Director to its extremity. I would be finishing support for Meet Mediaband so that it works glitchless.

Meet Mediaband works to a great extent on ScummVM's Director engine, but some loose ends need to be fixed.

Director Engine's Trello Board has a separate column for work to be done for Meet Mediaband. Accessible [here](#).

UnDo ME is a main section of Meet Mediaband. Its native working is shown [here](#).

The buttons UN and DO are not functional in ScummVM currently. Their intended functionality would be implemented.

The House Jam section (intended to work [here](#)) needs some fixes in rendering shapes and implementing the back button.

According to djsrv, there would be more bugs/glitches in Meet Mediaband that need fixing. As the motive of this task would be to play this target glitchless, this task largely covers fixing existing bugs in Meet Mediaband ScummVM.

Another thing to be implemented is the popup menu XObject in the xlibs. eientei provided me [this](#) README of popup XObj. The code with STUB for this XObj is in director/lingo/xlibs/popupmenuxobj.cpp (and its corresponding header)

Another issue I found is the usage of blocky dissolve transition instead of a smoother one (comparison here: [native](#) and [ScummVM](#)). The code which handles transition is [this](#). This is due to frame's

```
_transType = kTransDissolveBoxySquares
```

Instead of `kTransDissolvePixelsFast`, which is present in the target at some places. This is an inconsistency.

Also, the Macaroni Man sequence is not working in ScummVM. That would be another thing to fix.

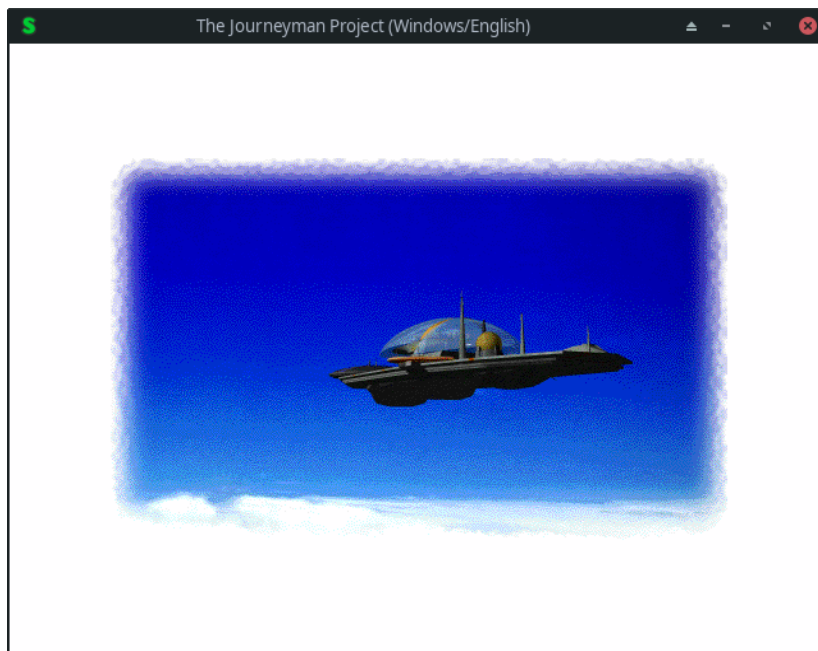
Along with the popup menu, the Credits screen, Puzzle, Day-Night, and Sunrise Sunset screens need to be working. ([Ref](#))

The Journeyman Project

ScummVM's Director engine has limited support for The Journeyman Project. I tested the Windows and Mac versions of the original game in ScummVM. The findings are given below:

Windows:

- Transitions are lacking in the intro slideshow. The same slides have the dissolve transition in Mac.



- Overflowing of text in messagebox. And when the messagebox is redrawn, the overflown text does not disappear.





This is an issue of sprites being rendered at the wrong places. There are several warnings on the terminal while running this target about wrong number of arguments for the messagebox, in which case a null argument is added. There are references to non-existent castmembers. This is the wrongly rendered castmember:

This is the right place for the description. But it also gets rendered at bbox: 25, 115, 195, 211, which is the messagebox.

A similar bug is also present in BioChip's description:



- The inventory is buggy. If a name spans two lines, the following line will refer to the other object, which is wrong.



Highlighting Biochip shows the Biochip Interface. But when the interface is highlighted (which shouldn't happen in the first place), the transport card is shown. Transport card can't be highlighted, while it must allow transport card to be highlighted.

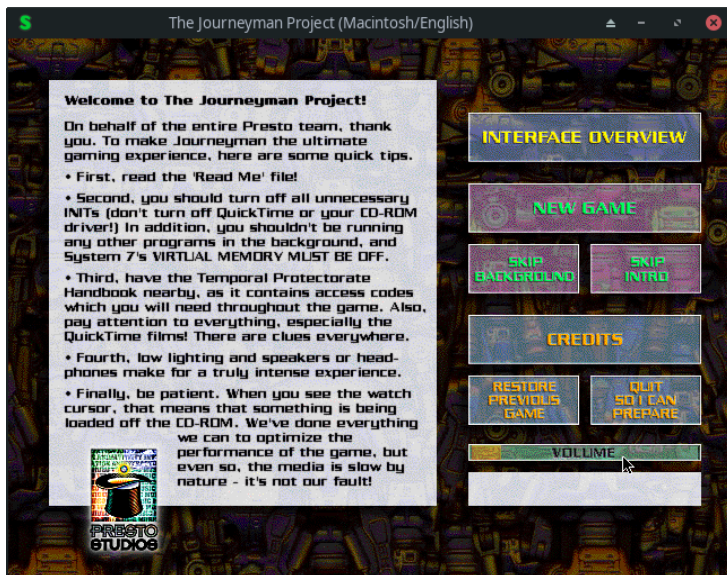
- **Major Game-Breaking Bug:** Target enters an infinite loop
Game freezes when the player bumps into a wall/door. The cursor movement is also very laggy, suggesting an infinite loop hogging resources. The following loop is repeating (got by using lingoexec debugflag)

```
[ 56]: Stack before:
[ 56]: c_intpush 8
[ 56]: Stack after: <8>
[ 58]: Stack before: <8>
[ 58]: c_theentitypush sprite, movieTime
Lingo::getTheEntity(sprite, 8, movieTime)
[ 58]: Stack after: <0>
[ 61]: Stack before: <0>
[ 61]: c_localpush "dur"
[ 61]: Stack after: <0> <30>
[ 63]: Stack before: <0> <30>
[ 63]: c_lt
[ 63]: Stack after: <1>
[ 64]: Stack before: <1>
[ 64]: c_jumpifz [ 72]
[ 64]: Stack after:
[ 66]: Stack before:
[ 66]: c_callcmd "updatestage", 0
call: updatestage()
[ 66]: Stack after:
[ 70]: Stack before:
[ 70]: c_jump [ 56]
[ 70]: Stack after:
```

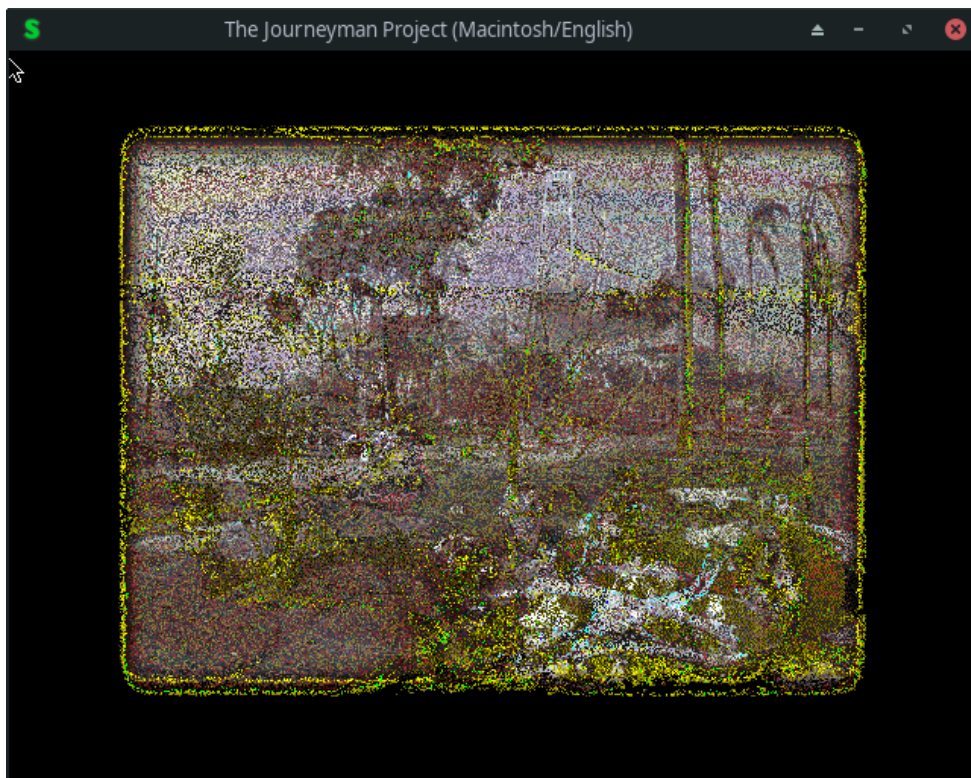

These were the bugs / possible areas of improvement I found during preliminary testing of the Journeyman Project for Windows

Mac:

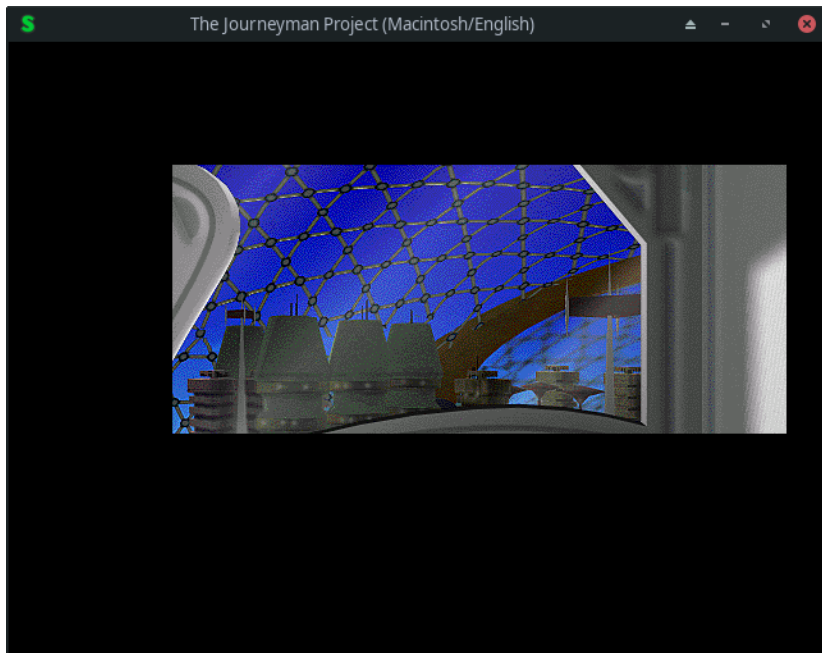
- The Volume functionality doesn't work in the main menu



- The sepia intro slideshow's color palette is messed up



- No Audio plays in the game in the Mac version after the main menu. Audio needs to be implemented into the engine.
- The HUD does not show up. So the game is unplayable.



```

WARNING: Lingo::resolveCastMember: reference to non-existent cast member: A Blank Castmember!
WARNING: Lingo::resolveCastMember: reference to non-existent cast member: MasterList!
WARNING: varAssign: Unknown member -1(???)!
WARNING: Lingo::resolveCastMember: reference to non-existent cast member: MasterList!
WARNING: evalChunkRef: Unknown member -1(???)!
WARNING: Lingo::resolveCastMember: reference to non-existent cast member: InventoryList!
WARNING: varAssign: Unknown member -1(???)!
WARNING: Lingo::resolveCastMember: reference to non-existent cast member: InventoryList!
WARNING: evalChunkRef: Unknown member -1(???)!
WARNING: BUILDBOT: Uncaught Lingo error: Call to undefined handler 'InterfaceInit'. Dropping 0 stack items!

```

These calls to non-existent castmembers are the HUD components that should have shown up.

- Skip Intro button doesn't skip the intro. Skip Intro and Skip background have the same effect in the Mac version of the game; they skip the sepia intro slideshow.

My tasks would be to resolve these issues and make The Journeyman Project (both Windows and Mac versions) largely playable. Fixing these issues would be the priority (especially the game-breaking ones)

Deliverables

Following is the set of deliverables that I would push:

- Implementing the current STUB code so that existing Workshop movies can work as expected.
- Make the UN and DO buttons work in Meet Mediaband's UNDO ME along with the back button
- Make the Macaroni Man sequence in Meet Mediaband work.
- Implementing the PopupMenu XObject to enable popup menus in the game. Clearing the Trello Board for Meet Mediaband.
- Fixing the stuck in a loop on the wall bump in The Journeyman Project (Win)
- Fix the Castmember rendering issue and Inventory issue in The Journeyman Project (Win)
- Fix the color palette for The Journeyman Project (Mac)
- Fix audio in The Journeyman Project (Mac)
- Fix the HUD and missing CastMembers in The Journeyman Project (Mac) and make it playable.
- Fixing the skip intro in The Journeyman Project (Mac) for intended functionality.
- Fixing other bugs which occur along the way in Meet Mediaband and The Journeyman Project (Windows and Mac) and make them largely playable.

Milestones

Following are the milestones of my GSoC, along with the reason for choosing them:

- Implementing STUB code and completing the Workshop Movies

- 
- This is an essential milestone as the STUB code affects all the targets which have to run and use that particular code. This would support all Lingo commands (Director 4) and the current STUB XObjects. This would resolve the missing Lingo Commands.
 - Completing glitchless support for Meet Mediaband
 - Meet Mediband is a target that uses the capabilities of Director 4 to its extremities. It is also a target on which a lot of work has been done previously by the dev team (djsrv worked extensively on it during the GSoC)
 - Completing support for The Journeyman Project (Win)
 - The Journeyman Project has some game-breaking glitches which are easy to encounter. It is also one of the prominent games using Director 4. So making it largely playable is one of the milestones of my GSoC.
 - Completing support for The Journeyman Project (Mac)
 - The Journeyman Project (Mac) is not playable in the current state of Director Engine. First, the castmembers need to be accessed correctly, and further fixes might be required. This is a significant milestone and, when done, would be a great addition to ScummVM.

Timeline

Following is the project timeline for my GsoC. I would be taking up the 350 hour format.

Date	Task
May 20, 2022 - June 12, 2022 Community Bonding	Discussion with sev and the rest of the director engine dev team about the project and continue working on the first milestone.
June 13, 2022 - June 26, 2022	Finalize work on STUB code. Start work on glitchless support of Meet Mediaband. Fix smaller issues like the dissolve effect and start working on UNDO ME.
June 27, 2022 - July 11, 2022	By this time I plan to complete implementation of UNDO ME, Macaroni Man
July 12, 2022 - July 25, 2022	Add PopupMenu to Meet Mediaband. This would have been implemented along with the xlibs, So this week might go in fixing other bugs I encounter in Meet Mediaband, and complete it, moving onto the next milestone.
July 26, 2022 - August 8, 2022	Fix the wall bump glitch in The Journey Project Windows to make it more playable. If complete mid-week, move onto the castmember glitch fixing.
July 10, 2022 - July 23, 2022	Debug and fix the castmember glitch and make the game largely playable. Do small fixes (like the slide transitions) and other bugs which would have come up during playtesting

July 24, 2022 - August 6, 2022	Testing the changes in The Journeyman Project (Win). Fixing anymore bugs that are encountered. These 2 weeks can also accommodate some work which took longer. Complete The Journeyman Project (Win). Mid Evaluation
August 7, 2022 - August 20, 2022	Start work on The Journeyman Project (Mac). Fix the Skip Intro button and look into the HUD of The Journeyman Project (Mac) (Basically work on the missing castmembers)
August 21, 2022 - September 12, 2022	Fix the HUD of The Journeyman Project by implementing the castmembers, making the game playable (able to make progress)
September 13, 2022 - October 2, 2022	Add sounds in The Journeyman Project (Mac), making it largely playable. I would be having my midsem exams in the third week of September.
October 3, 2022 - October 23, 2022	This time is for fixing any unexpected blockers in the previous targets (most probably for The Journeyman Project (Mac) as I haven't been able to find any glitches due to the game being unplayable).
October 24, 2022 - November 13, 2022	In the last phase of GSoC, I would complete The Journeyman Project (Mac) . This time is kept for extensive testing and fixing any other bugs

	encountered. I would have my Endsem exams during this period
November 14, 2022 - November 21, 2022	Final Evaluation. Wrapping up the work done this summer and documenting it.

I have kept ample time to mitigate any unexpected blockers. The blockers can mainly pop up from The Journeyman Project (Mac), as the target is not playable in its current state.

I have also mentioned the schedule for my university exams and have given a bigger timeframe to those tasks which coincide with my exams.

If everything goes better than expected, I would move onto my stretch goal - Adding support for **Director 5** in ScummVM. I won't be completing this in the GSoC timeline, but would rather initiate work on it and continue developing along with the dev team Post-GSoC too. Director 5 support is in very initial stages in ScummVM and I have not elaborated upon this in the proposal, so this would be discussed with the dev team then. [Here](#) is a doc that djsrv shared, containing the TODO tasks for Director 5.

What makes me the best person to work on this project?

Following are the reasons which justify why I am the best person to work on this project:

- **Good knowledge of C++:** I primarily code in C++ and am pretty familiar with its internals and the Object-Oriented paradigm of C++. I am also familiar with build tools, Make, and GDB to aid me in programming and debugging.
- **Familiarity with codebase:** I have been with the community and contributed to ScummVM since January 2022. Most of the time, I have

been skimming the code of various components, understanding how engines work (both VMs and hardcoded ones)

- **Contributing to the code:** I have already made several contributions to fixing existing bugs and adding functionality. The last couple of PRs were part of my First Milestone, and I am working on completing my first milestone.
- **Availability of Director Games:** I have a couple of Director games with me - Meet Mediaband, Spaceship Warlock (Windows and Mac), Chop Suey, and The Journeyman Project (Windows and Mac). I also have Sheepshaver and Basilisk II for running Mac Director games. Along with plenty of demo targets (Like Microsoft Musical Instruments, Apple Tour of Macintosh Portable), I have plenty of games and targets to test out.
- **Knowledge of the Director Engine:** I have various reading resources for the Macromedia Director (The Lingo Dictionary, The Studio Manual, Director tips and tricks, The Interactivity Manual). So seeking references or trying things out in Director Engine would be easy for me.
- **Prior coding experience in a team:** I am a part of SDS Labs, a campus group at IITR focused on software development and CTFs. Working on projects undertaken by SDS Labs, I know how to work in a group while developing a piece of software, effective use of version control systems, and the importance of communication.
- I have already elaborated about my university and my projects in the About Me section.

What attracts me to ScummVM and the Director Engine

What I like about ScummVM

The ScummVM project seems to have a diverse set of tasks. There are a lot of engines in there, all needing a lot of things: The main loop (The VM part), Audio System, Graphics system, Inputs, etc. There are engines with TTS



support and engines that support 3D graphics. We can be working on the debugger, the game detection system, the audio or graphics system, or contribute to an engine. This variety of work was also visible in previous GSoC cohorts' projects.

I am passionate about C++ projects (I worked on [Rootex](#) - an advanced 3D game engine written mainly in C++). I am well versed in the language's fundamentals and its Object-Oriented Programming paradigm. All the components of every engine are written from scratch.

The ScummVM dev team acknowledges the work done by GSoC students, and students have initiated/contributed to projects taken very far by the team. Also, many GSoC students continue their contributions and become members of the dev team. Work done by the contributors and students is respected in the ScummVM community.

The community is brilliant and extremely helpful. Communication and doubts are appreciated, and the replies are prompt (because almost none of the devs are working on ScummVM full-time and are dispersed across various time zones)

What attracts me to the Director Engine specifically?

ScummVM has 92 game engines and supports over 6000 games. [Source](#)

Extending the director engine would extend ScummVM's list of supported games. It would triple the number of games supported currently.

Much work has been done on the engine by the dev team and the previous students, and I would like to build upon it and contribute significantly to the Director Engine. Director Engine is also being developed very actively by the team, and the communication on the discord channel is excellent.

Benefits to the ScummVM community

With my contributions, I would be bringing the Director Engine to a complete state and making certain games work glitchless on the engine. Extending the engine's capabilities would also remove glitches from other Director games.

The Director Engine would be adding support for a lot of games to ScummVM, increasing the roster of games that ScummVM can run.

I am also interested in contributing to the Director Engine Post-GSoC (I would have a lot of knowledge of the codebase by then) to make it good enough to be released, fix bugs, or add functionality. I would also like to work on other components of ScummVM (hoping to be a part of the dev team)

I also do reverse engineering challenges in my spare time, so I can take up a game reversing project Post-GSoC.

Timezone

I live in the Indian Standard Time (GMT +0530). My waking hours are 11 AM IST to 4 AM IST (6:30 AM GMT to 10:30 PM GMT) and prefer to work from 6 PM to 4 AM IST (12:30 PM to 10:30 PM GMT)

Development Environment and Devices

To build and code ScummVM, I use the following systems:

- **Windows**
 - Visual Studio 2019 IDE for coding, building and debugging
- **Manjaro Linux, Ubuntu 20.04, Mac Mini**
 - JetBrains CLion IDE for coding, building and debugging